

Lecture(17)

Chapter(5)

Counting

- Basic counting principles.
- Factorial notation.
- Binomial Coefficients and Pascal's Triangle.

One, two, three, we're...

Counting

- We must count objects to solve many different types of problems. For instance, counting is used to determine the complexity of algorithms.
- Counting is also required to determine whether there are enough telephone numbers or Internet protocol addresses to meet demand.
- Furthermore, counting technique are used extensively when probabilities of events are computed.

Basic Counting Principles

Counting problems are of the following kind:

“**How many** different 8-letter passwords are there?”

“**How many** possible ways are there to pick 11 soccer players out of a 20-player team?”

Most importantly, counting is the basis for computing **probabilities of discrete events**.

(“What is the probability of winning the lottery?”)

Basic Counting Principles

The sum rule:

If a task can be done in n_1 ways and a second task in n_2 ways, and if these two tasks cannot be done at the same time, then there are $n_1 + n_2$ ways to do either task.

Example:

1- The department will award a free computer to either a CS student or a CS professor. How many different choices are there, if there are 530 students and 15 professors?

There are $530 + 15 = 545$ choices.

Basic Counting Principles

Generalized sum rule:

- If we have tasks $T_1, T_2, \dots, T_m$ that can be done in $n_1, n_2, \dots, n_m$ ways, respectively, and no two of these tasks can be done at the same time, then there are $n_1 + n_2 + \dots + n_m$ ways to do one of these tasks.

Basic Counting Principles

Examples:

2- Suppose E is the event of choosing a prime number less than 10, and F is the event of choosing an even number less than 10. Then E can occur in four ways $\{2, 3, 5, 7\}$, and F can occur in 4 ways $\{2, 4, 6, 8\}$. However E or F can not occur in $4 + 4 = 8$ ways since 2 is both a prime number less than 10 and even less than 10. In fact, E or F can occur in only $4 + 4 - 1 = 7$ ways.

3- Suppose E is the event of choosing a prime number between 10 and 20, and suppose F is the event of choosing an even number between 10 and 20. Then E can occur in 4 ways $\{11, 13, 17, 19\}$, and F can occur in 4 ways $\{12, 14, 16, 18\}$. Then E or F can occur in $4 + 4 = 8$ ways since now none of the even numbers is prime.

4- Suppose that either a member of the mathematics faculty or a student who is mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors?

Solution: From the sum rule there are $37 + 83 = 120$ possible ways to pick this representative.

5- A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. How many possible projects are there to choose from?

Solution: The student can choose a project from the first list in 23 ways, from the second list in 15 ways, and from the third list in 19 ways. Hence, there are $23 + 15 + 19 = 57$ projects to choose from.

Basic Counting Principles

The product rule:

Suppose that a procedure can be broken down into two successive tasks. If there are n_1 ways to do the first task and n_2 ways to do the second task after the first task has been done, then there are $n_1 \cdot n_2$ ways to do the procedure.

Example:

1- How many different license plates are there that containing exactly three English letters ?

There are 26 possibilities to pick the first letter, then 26 possibilities for the second one, and 26 for the last one. So there are $26 \cdot 26 \cdot 26 = 17576$ different license plates.

Basic Counting Principles

Generalized product rule:

If we have a procedure consisting of sequential tasks $T_1, T_2, \dots, T_m$ that can be done in $n_1, n_2, \dots, n_m$ ways, respectively, then there are $n_1 \cdot n_2 \dots \cdot n_m$ ways to carry out the procedure.

Basic Counting Principles

Examples:

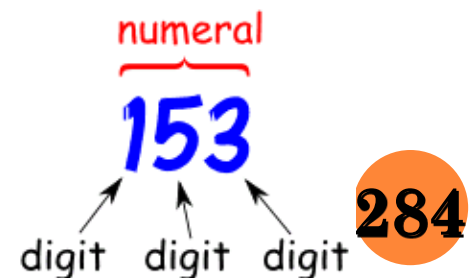
2- Suppose license plate contains two letters followed by three digits with the first digit not zero. How many different license plates can be printed?

Solution: Each letter can be printed in 26 different ways, the first digit in 9 ways and each of the other two digits in 10 ways. Hence $26 \cdot 26 \cdot 9 \cdot 10 \cdot 10 = 608400$ different plates can be printed.

3- In how many ways can an organization containing 26 members elect a president, treasurer, and secretary (assuming no person is elected to more than one position)?

Solution: The president can be elected in 26 different ways; following this, the treasurer can be elected in 25 different ways, and, following this, the secretary can be elected in 24 different ways. Thus, by the above principle of counting, there are $26 \cdot 25 \cdot 24 = 15600$ different ways.

- A digit is a single symbol used to make numeral.
- 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9 are the ten digits we use in everyday numerals.



Sum and product rule principle

The sum and product rules can also be phrased in terms of **set theory**.

Sum rule: Let $A_1, A_2, \dots, A_m$ be disjoint sets. Then the number of ways to choose any element from one of these sets is

$$|A_1 \cup A_2 \cup \dots \cup A_m| = |A_1| + |A_2| + \dots + |A_m|.$$

Product rule: Let $A_1, A_2, \dots, A_m$ be finite sets. Then the number of ways to choose one element from each set in the order $A_1, A_2, \dots, A_m$ is

$$|A_1 \times A_2 \times \dots \times A_m| = |A_1| \times |A_2| \times \dots \times |A_m|.$$

Factorial notation

The factorial of nonnegative integer **n** denoted by **n!**, is the product of all positive integer less than or equal to n.

$$3 \times 2 \times 1 = 6$$

Written using factorial notation



Which means

3!

Pronounced as
“three factorial”

Factorial notation

- The product of positive integers from 1 to n inclusive is denoted by $n!$ (read “ n factorial”):

$$\text{In general } n! = n(n-1)(n-2)(n-3) \dots (3)(2)(1)$$

- In other words, $n!$ is defined by

$$n! = \begin{cases} n.(n-1)! & \text{if } n > 2 \\ 1 & \text{if } n = 0 \text{ or } 1 \end{cases}$$

Illustration:

$$2! = 2.1 = 2,$$

$$3! = 3.2.1 = 6$$

$$4! = 4.3.2.1 = 24$$

$$5! = 5.4! = 5.24 = 120$$

$$6! = 6.5! = 6.120 = 720$$

Factorial notation

a) Simplify $\frac{n!}{(n-2)!} = \frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}}$

b) Simplify $\frac{8!}{6!} = \frac{8 \times 7 \times \cancel{6!}}{\cancel{6!}}$

c) Express $10 \times 9 \times 8 \times 7$ as a factorial.

$$= \frac{10!}{6!}$$

Binomial Coefficients

- The symbol $\binom{n}{r}$ (read “nCr”), where r and n are positive integers with $r \leq n$, is defined as

$$\binom{n}{r} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)\dots 3.2.1}$$
$$\Rightarrow \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- We have the following important relation :

$$\binom{n}{r} = \binom{n}{n-r}$$

Binomial Coefficients

Illustration(1):

$$\begin{aligned}\binom{8}{2} &= \frac{8 \cdot 7}{2 \cdot 1} = 28, & \binom{9}{4} &= \frac{9 \cdot 8 \cdot 7 \cdot 6}{4 \cdot 3 \cdot 2 \cdot 1} = 126 \\ \binom{12}{5} &= \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 792, & \binom{10}{3} &= \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120\end{aligned}$$

Illustration(2):

$$\binom{10}{7} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4}{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 120$$

- On the other hand, $10 - 7 = 3$ and so we can also compute as follows:

$$\binom{10}{7} = \binom{10}{3} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 120$$

- **Observe that the second method saves space and time**

Useful facts about power series

- The numbers $\binom{n}{r}$ are called the binomial coefficients since they appear as the coefficients in the expansion of $(a + b)^n$.

Theorem:

Let x be variables, and let n be a nonnegative integer. Then

$$\begin{aligned}(a + b)^n &= \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \\ &= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{n-1} a b^{n-1} + \binom{n}{n} b^n.\end{aligned}$$

Binomial Coefficients and Pascal's Triangle

Examples:

(a) $(x + 2)^4 =$

$$= \binom{4}{0}x^4 + \binom{4}{1}x^3 \cdot 2 + \binom{4}{2}x^2 \cdot 2^2 + \binom{4}{3}x \cdot 2^3 + \binom{4}{4} \cdot 2^4$$

$$x^4 + 8x^3 + 24x^2 + 32x + 16.$$

(b) $(x + 3)^5 =$

$$= \binom{5}{0}x^5 + \binom{5}{1}x^4 \cdot 3 + \binom{5}{2}x^3 \cdot 3^2 + \binom{5}{3}x^2 \cdot 3^3 + \binom{5}{4}x \cdot 3^4 + \binom{5}{5} \cdot 3^5$$

$$x^5 + 15x^4 + 90x^3 + 270x^2 + 405x + 243.$$

Binomial Coefficients and Pascal's Triangle

- The coefficients of the successive powers of $a + b$ can be arranged in a triangular array of numbers, called Pascal's triangle, as pictured in next slide. The numbers in Pascal's triangle have the following properties:
 1. The first number and the last number in each row is 1.
 2. Every other number in the array can be obtained by adding the two numbers appearing directly above it.

Binomial Coefficients and Pascal's Triangle

				1						ROW 0
				1		1				ROW 1
			1	2		1				ROW 2
		1	3		3	1				ROW 3
		1	4	6		4	1			ROW 4
	1	5	10	10		5	1			ROW 5
	1	6	15	20	15		6	1		ROW 6
	1	7	21	35	35	21	7	1		ROW 7
1	8	28	56	70	56	28	8	1		ROW 8
1	9	36	84	126	126	84	36	9	1	ROW 9



Yeah! Its finished.

Binomial Coefficients and Pascal's Triangle

Pascal's triangle

$$(a + b)^0 = 1$$

$$(a + b)^1 = a + b$$

$$(a + b)^2 = a^2 + 2a b + b^2$$

$$(a + b)^3 = a^3 + 3a^2b + 3a b^2 + b^3$$

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4a b^3 + b^4$$

$$(a + b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5a b^4 + b^5$$

$$(a + b)^6 = a^6 + 6a^5b + 15a^4b^2 + 20a^3b^3 + 15a^2b^4 + 6ab^5 + b^6$$